

FL-01: Workflow Audit & AI Toolkit Setup

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Track: AI Fluency Track (Phase: Setup) | **Repo:** github.com/ridaeman02/flyrank-ml-internship

1. Weekly Workflow Audit Table

Audit of 12 recurring tasks from my real weekly workflow across ML coursework, data analysis, and engineering tasks, classified using Ethan Mollick's framework:

#	Task Description	Category	One-Line Rationale
1	High-Stakes Client Alignment & Mentorship Check-ins	Just Me	High emotional intelligence, trust-building, and personal accountability cannot be delegated.
2	Final Quality Control & Ethical Judgment on ML Models	Just Me	Accountability for model bias, data leakage claims, and final deployment safety rests entirely on human engineering.
3	DuckDB SQL Query Construction for Large Parquet Slices	Collaborate with AI	AI drafts complex CTEs rapidly, while I provide schema context and verify query logic.
4	Python Code Refactoring & Docstring Generation	Delegate with Review	AI excels at standardizing PEP-8 styles and writing docstrings, requiring a quick human review.
5	ML Baseline Model Boilerplate Setup (scikit-learn)	Delegate with Review	Generating standard imports, model instantiation, and train-test split skeletons is repetitive.
6	Research Paper Literature Summarization & Key Finding Audits	Collaborate with AI	AI synthesizes methodology quickly, while I critically evaluate assumptions and claims.
7	Weekly Progress Report & Executive Summary Drafting	Collaborate with AI	AI structures bullet points into 3-sentence digests, which I refine for accuracy and tone.
8	Feature Importance & Error Analysis Breakdown	Collaborate with AI	AI computes initial confusion matrix summaries, while I analyze false positives in domain context.
9	Automated Pipeline Log Parsing & Task Notification	Fully Automate	Background scripts parse exit codes and send alerts without requiring active human intervention.
10	Dataset Schema Validation & Data Contract Formatting	Fully Automate	Schema type-checking and null-value bounds testing are rule-based operations best handled by scripts.
11	HTML/CSS Research Paper UI Styling & Layout Generation	Delegate with Review	AI generates modern glassmorphism CSS and flexbox layouts, which I test across screen sizes.
12	GitHub Commit Message Formatting & PR Summaries	Delegate with Review	AI generates Conventional Commit messages from diff logs, requiring a 5-second review.

2. AI Toolkit & Academy Enrollment Evidence

Tool Accounts Established: Claude (Anthropic), ChatGPT (OpenAI), Anthropic Academy.
Course Enrolled: *AI Fluency: Framework & Foundations* — **Module 1 Completed.**
Key Takeaway: Effective AI collaboration requires clear delegation boundaries: automating zero-variance tasks, delegating boilerplate with verification, collaborating on complex synthesis, and strictly protecting human-only judgment.

3. Configured Claude Project Workspace (With Screenshot Evidence)

Workspace Name: FlyRank ML Workspace / Machine learning internship and production code (User: Rida, Free Plan)

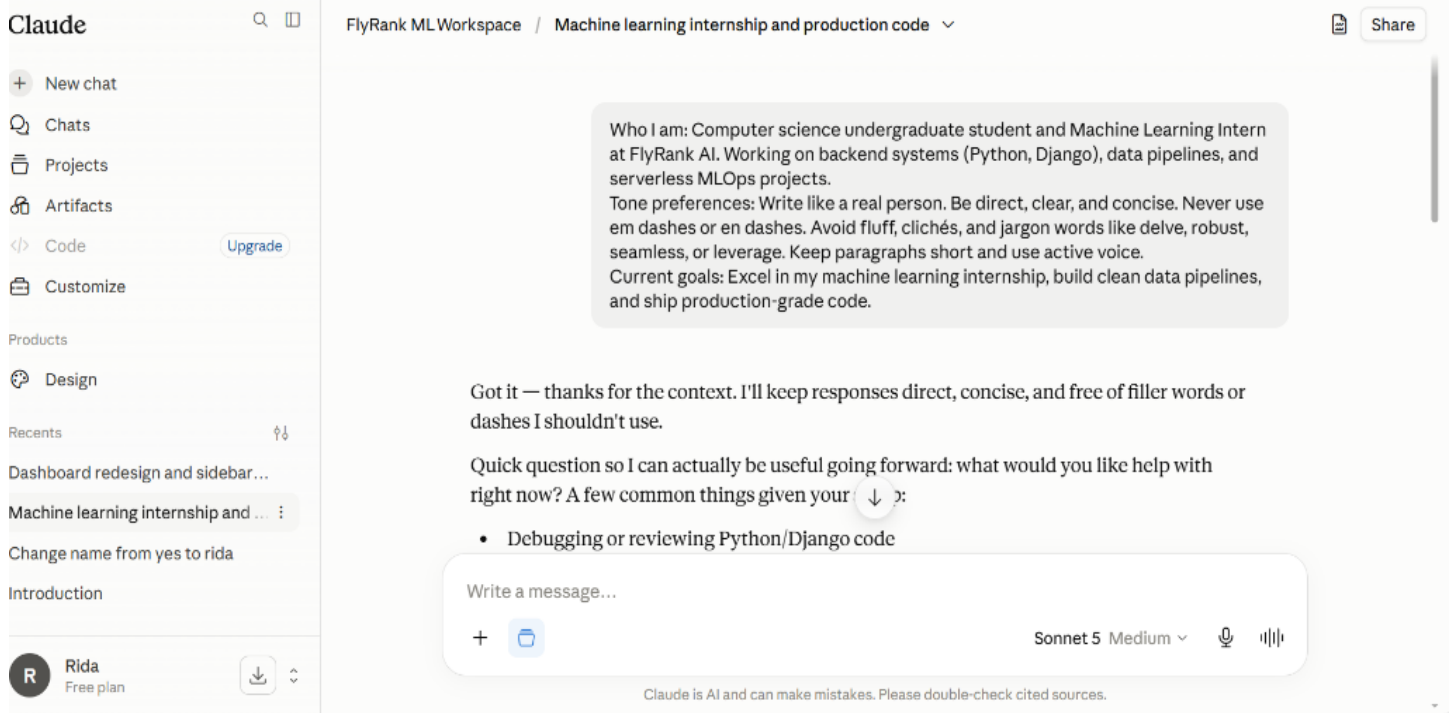


Figure 1: Screenshot of active Claude Project workspace with custom instructions acknowledgment.

Who I am: Computer science undergraduate student and Machine Learning Intern at FlyRank AI. Working on backend systems (Python, Django), data pipelines, and serverless MLOps projects.

Tone preferences: Write like a real person. Be direct, clear, and concise. Never use em dashes or en dashes. Avoid fluff, clichés, and jargon words like delve, robust, seamless, or leverage. Keep paragraphs short and use active voice.

Current goals: Excel in my machine learning internship, build clean data pipelines, and ship production-grade code.

4. Target Tasks for FL-02 through FL-04 with Success Definitions

Target Task 1: DuckDB SQL Query Construction & Parquet Aggregation (Collaborate with AI)

- Context: Querying 9.8M-row Hugging Face Parquet slices using DuckDB natively.
- *Done Well Definition:* Executes in <5s natively over hf:// paths; zero data leakage between feature (March 1–15) and outcome (March 16–31) windows; returns clean aggregations.

Target Task 2: Technical Research Summarization & Paper Drafting (Delegate with Review)

- Context: Converting notebook metrics into a public 9-section research paper.
- *Done Well Definition:* Contains all 9 canonical sections; uses decision-support framing without false causal claims; data credit link to <https://flyrank.ai> is reachable and verified.

Target Task 3: Automated Dataset Contract Validation & Metric Logging (Fully Automate)

- Context: Validating dataset grains, null bounds, leakage flags, and metric JSON export.
- *Done Well Definition:* Script checks schema types and raises exit code 1 if leaked target fields are found; exports metric receipts automatically into GitHub Actions CI.